

UMA MIRACLE IDIKA

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EDUCATION.

Federal University of Technology Owerri, Imo State, Nigeria

Sept. 2019 - Oct. 2025

Bachelor of Engineering in Petroleum and Gas Engineering

Final Grade: First Class Honors (**Rank: 1/130**)

CGPA: 4.65/5.00

Scholaro CGPA: 3.84/4.00

Relevant Courses: Reservoir Modelling and Simulation, Drilling Technology, Production Engineering, Statistical Thermodynamics, Enhanced Oil Recovery, Transport Phenomena, Computational Fluid Dynamics.

RESEARCH EXPERIENCE.

Research Support, Schlumberger, Colorado, United States of America.

June 2025 - Present

Bridging Arps Decline Theory & Deep Learning Physics-Constrained N-N Framework: 864 Tight-Gas Wells in the Piceance Basin

- Developed a physics-constrained hybrid neural network framework to forecast multi-horizon gas production rates in unconventional gas plays, overcoming the limitations of fixed-form decline curve models.
- Benchmarked the model against classical DCA variants (Arps, Duong, SE) and deep sequence forecasters across 864 Mancos Shale wells, with well-specific decline family selection and physics-penalized residual correction.
- Achieved a median RMSE of 9.63 MCF/day, outperforming all benchmarks in 95% of wells with a 4.9% mean RMSE reduction, confirming that physics regularization produces more accurate forecasts than unconstrained DL models.

Research Team Lead, Society of Petroleum Engineers, FUTO

Jan. 2024 - Apr. 2026

Led the SPE FUTO Research Team (Energy Challenge & Paper Contest) on the exhibitions:

Evaluation of Palm Oil Mill and Glycerol Derived Nutrients for Biogenic CH₄ Production in Depleted Gas Reservoirs

- Stimulated indigenous methanogenic archaea in depleted Niger Delta gas reservoirs using a hybrid POME-glycerol formulation, with trace metals and nitrate dosing to enhance methanogenesis and suppress reservoir souring.
- Modelled microbial growth kinetics across three well scenarios using a Monod kinetics-based framework parameterized with Agbada Formation data, under varying temperature, pressure, pH, and salinity conditions.
- Achieved a 25–33% increase in microbial growth kinetics at optimized pH 6.8-7.2, validating agro-waste-driven stimulation as a low-emission, cost-effective strategy for extending depleted gas field life.
- Awarded a 1st place National Paper Contest finish by SPE, Nigeria Council, 2026.

Biogas Generation Using Agro-Waste Materials: An Experimental and Machine Learning Approach

- Developed a novel approach to maximize Biogas yield from Agro-Waste via the alteration of the Carbon:Nitrogen ratios
- Developed a novel Acid Gas adsorption system from palm burnt ash, achieving an average adsorption purity of 98.2%
- Developed a Recurrent Neural Network (RNN)-based produced gas simulator, that provides real time insights into the expected biogas yield (ltr/day) from 7 input real-time sensor parameters.
- Awarded the 4th best energy innovation in Nigeria, 2024, by SPE, Nigeria Council.

Undergraduate Research Assistant under Dr. N.U. Okereke

Sept. 2023 - Feb. 2025

Predicting Oil Sludge Formation During Crude Oil Production In The Niger Delta Using Machine Learning Models

- Conducted a scientific investigation into sludge accumulation as a critical flow assurance hurdle in the Niger Delta.
- Assessed the efficacy of 5 ML regression frameworks using 144 crude oil data points to predict sludge fallout during production, with PolySVR achieving superior predictive accuracy ($R^2 = 0.976$; RMSE = 0.00876 g/g) via nested cross-validation. and establishing wax concentration as the primary driver for sludge precipitation [[link](#)]

PUBLICATIONS.

- **U.I., Miracle**, Dr. N. Okereke, Kerunwa, Dr. E. Odoh (2026). Predicting Oil Sludge Formation During Crude Oil Production in the Niger Delta using Machine Learning Models. *OnePetro* (under peer-review)
- E.C., Obasi, **U.I., Miracle** (2026). Bridging Arps Decline Theory & Deep Learning Physics-Constrained Neural Network Framework across 864 Tight-Gas Wells in the Piceance Basin. *OnePetro* (under peer-review)
- O. Cherish, **U.I., Miracle** (2026). Evaluation Of Palm Oil Mill Effluent (POME) And Glycerol-Derived Nutrients For Stimulating Biogenic Methane Production In Depleted Gas Reservoirs. *OnePetro* (under peer-review)

TEACHING EXPERIENCE.

Machine Learning Instructor

Apr. 2021 - Feb. 2025

Society of Petroleum Engineers, FUTO., Nigeria

- Tutored over 100 students on the practical applications of ML and Deep Learning in Petroleum Engineering.

Academic Tutor

Oct. 2020 - Dec. 2024

Petroleum Engineering Department, FUTO, Nigeria.

- Prepared academic materials and organized tutorials on Reservoir Modelling and Simulation, Fluid Dynamics, Heat and Mass Transfer, etc, for a class of over 150 students, recording an average success rate of over 85% per semester.

Academic Coordinator

Sept. 2022- Dec. 2024

Nigerian Fellowship of Evangelical Students, FUTU, Nigeria.

- Tutored, and mentored over 100 undergraduate students, achieving an annual average academic success rate of over 96%.

PROFESSIONAL EXPERIENCE.

Graduate Engineering Intern

Oct. 2025 – Present

CMES-OMS Petroleum Development Company, Ikoyi, Lagos, Nigeria..

- Conducted Well Planning Drillability and Anti-Collision Assessments for wells NAB-3 & NAB-4, ensuring safe and optimized well trajectories in compliance with industry standards. Presented findings in a high stake environment [[link](#)]

Machine Learning Engineer.

July. 2025 – Dec. 2025

LearnFactory Nigeria, Abia State, Nigeria.

- Modeled Buckley Leverett water-flooding Improved Oil Recovery and Van-Hurst Aquifer models using Python scripts.
- Developed time-series Deep Learning frameworks and engineered features from production and reservoir datasets to classify well behavior, forecast production trends, and support reservoir performance insights using CNN. [[link](#)]

Production Engineering Intern.

Jan. 2024 – June 2024

Seplat Energy Plc (OML 53), Imo State, Nigeria.

- Resolved a 5-year sludge precipitation problem at OML 53; the solution achieved >80% reduction in sludge occurrence and equipment fouling, alongside a 0.15 g/g decrease in sludge mass precipitation over a 6-month monitoring period.
- Analyzed daily production data, and conducted crude oil laboratory analyses (BS&W, API gravity, sand & wax content) across 36 oil wells, synthesizing results into production charts that provided real-time visibility into well performance.

Drilling Fluids Laboratory Intern.

Sept. 2022 - Dec. 2022 | July 2023 - Oct. 2023

Federal University of Technology Owerri, Imo State, Nigeria.

- Engineered and evaluated bio-diesel via transesterification from locally sourced Palm Kernel Oil and Vegetable Oil, optimising reaction conditions to maximise FAME yield and meet standard SOBM quality benchmarks.
- Formulated Environmentally Friendly Synthetic Oil-Based Mud (SOBM) using the produced bio-diesel as base fluid; evaluated fit-for-purpose performance across its thixotropic properties, fluid loss, and filter cake quality.
- Developed and evaluated HPHT cement slurries using both locally sourced and conventional additives; compared their influence on thickening time, compressive strength, and thixotropic behaviour under HPHT conditions.

CONFERENCES ATTENDED/PRESENTED AT.

- Nigeria Annual International Conference and Exhibition, 2025.
- SPE Africa Annual Students Congress 2025: Building the energy workforce of tomorrow.
- SPE S103 Annual Students Energy Conference 2024: From campus to career advancement.
- SPENC Students Technical Symposium and Exhibition 2024: Securing the future of the energy industry.

SELECTED LEADERSHIP AND VOLUNTEERING SERVICES.

United Leaders Global Ambassador.

Jan. 2026 - Present

- Appointed the global ambassador and advocate for Sickle Cell warriors across the globe by the United Leaders Global, where my team and I pioneered campaigns for Sickle Cell awareness, hospital visits and charity fundings.

Research Team Lead, Society of Petroleum Engineers, FUTU.

Dec. 2024 - Mar. 2026

- Led the SPE FUTU Research Team (Energy Challenge and Paper Contest), managed the organization of the projects exhibited, and the project presentations in collaboration with the Petroleum Engineering Professors.

Tube Mentorship Program, TMP 4.0.

June 2025 - Dec. 2025

- Led a team of 7 through an intensive 6-month technical research program, where our research focused on proposing sustainable solutions and strategies for ethical and sustainable AI integration in Ghana, ultimately being named the overall best performing team out of 17 teams in the cohort. [[link](#)]

Nigerian Fellowship of Evangelical Students' (NIFES) Academic Coordinator

Sept. 2022 - Aug. 2024

- Organized programs geared towards improving the academic lifestyle of students such as effective time management, life after school, balancing christian and social life with academics, etc.

SKILLS AND PROFICIENCIES.

- Scientific Programming: Proficient in Python, MATLAB, & C++ for scientific programming, data science, & analytics.
- Machine Learning Expertise: Skilled in Supervised, Unsupervised, Semi-Supervised Learning, and Deep Learning.
- Software Tools: Competent in Microsoft Office, Google Suites, MLOps, PROSPER, GIS, and ASPEN HYSYS.
- Technical and Laboratory: Well Testing, Drilling Fluids Study, Crude Oil Laboratory Analysis, Core Sample Analysis.
- Soft Skills: Excellent Communication, Leadership & Team Player, Organizational Skills, Time & Project Management.

SELECTED AWARDS & HONORS.

Academic Achievements

- Overall Best Graduating Student, Petroleum Engineering [\[link\]](#)
- Perfect CGPA in Departmental Courses: **5.00/5.00**.
- Departmental Research Award: Best student with more than 94% passed in final year undergraduate research work.

Competition Achievements

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| • SPE Students Technical Symposium and Exhibition: 1 st Place, National Paper Contest. | <i>Mar. 2026</i> |
| • SPE Africa DSEATS Datathon: Best Data-Driven Production Optimization solution (1 st /318 solutions) | <i>July 2025</i> |
| • SAASC 2025: Top Individual Performer, Africa Petrobowl Tournament. [link] | <i>Apr. 2025</i> |
| • SPE S103 Annual Students Conference: 2 nd place Regional PetroQuiz Tournament. (2 nd /60 participants) | <i>Dec. 2024</i> |
| • SPE Students Technical Symposium and Exhibition: 4 th Best Energy Innovation in Nigeria, 2024. [link] | <i>Mar. 2024</i> |

Scholarships

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| • TechRise 2.0 AI Scholar – Emerging Leader in AI 2025, Learn Factory, Nigeria. | <i>July 2025</i> |
| • Shekinah Scholarship - Outstanding Academic Performance | <i>Jan 2013</i> |

CERTIFICATIONS.

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| • AI & ML for Geospatial Analysis - Udemy | • National Skills Qualification in AI - CPRCN. |
| • IT Specialist in AI - Certiport, Pearson Vue | • OSHA UK HSE Supervisor - Level 1, 2 & 3 |

PROFESSIONAL REFERENCES**1. Dr. Ndubuisi Okereke**

- Title: Senior Lecturer and Faculty Adviser.
- Institution: Federal University of Technology, Owerri.
- Relationship: Undergraduate Thesis Supervisor and Class Adviser.
- Email: ndubuisi.okereke@futo.edu.ng
- Phone: +2347013927935
- LinkedIn: [Link](#)

2. Engr. James Uchendu

- Title: Chief Operating Officer.
- Organization: CMES-OMS Petroleum Development Company.
- Relationship: Boss and Career Coach.
- Email: james.uchendu@ept.ng
- Phone: +2348026486136
- LinkedIn: [Link](#)

3. Precious Okiemute Eku

- Title: Senior Advisor, HRBP, Seplat Energy Plc - Nigeria.
- Organization: Seplat Energy Plc
- Relationship: Industry-based Supervisor during Undergraduate Internship.
- Email: OEku@seplatenergy.com
- Phone: +2348067156229
- LinkedIn: [Link](#)